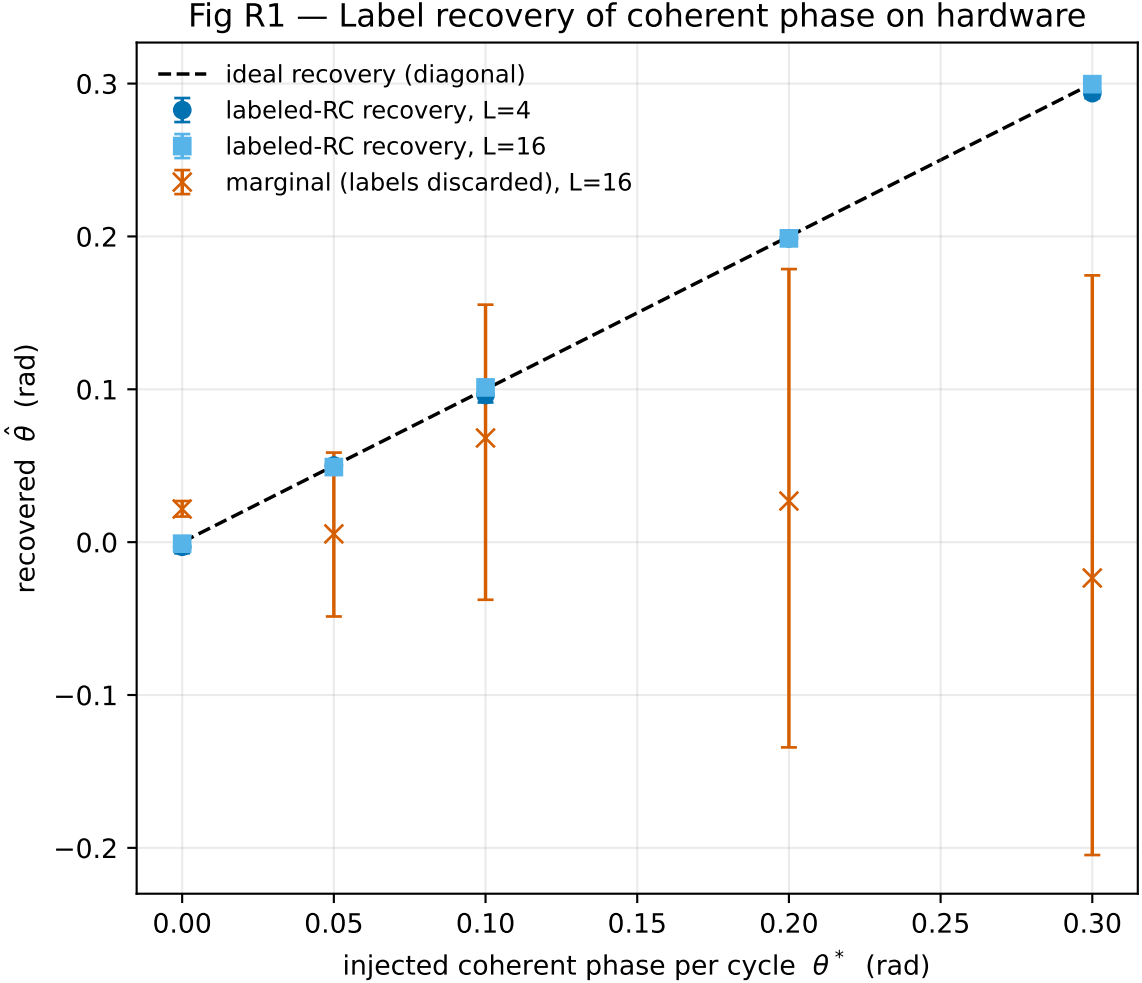
Fig R1 — Label recovery of coherent phase on hardware
ideal recovery (diagonal)
labeled-RC recovery, L=4
labeled-RC recovery, L=16
marginal (labels discarded), L=16
recovered $\hat{\theta}$ (rad)
0.3
0.2
0.1
0.0
−0.1
−0.2
0.00
0.05
0.10
0.15
0.20
0.25
0.30
injected coherent phase per cycle θ^* (rad)
backend=ibm_marrakesh, qubit=2, date=2026-07-21, shots=3000/circuit, 50 twirl instances/point